J. Dairy Sci. 106:4874–4895
https://doi.org/10.3168/jds.2022-22447
© 2023, The Authors. Published by Elsevier Inc. and Fass Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Epidemiology of bovine colostrum production in New York Holstein herds: Cow, management, and environmental factors

T. A. Westhoff,[1] S. J. Womack,[1] T. R. Overton,[2] C. M. Ryan,[2] and S. Mann[1]*
[1]Department of Population Medicine and Diagnostic Sciences, College of Veterinary Medicine, Cornell University, Ithaca, NY 14853
[2]Department of Animal Science, College of Agriculture and Life Sciences, Cornell University, Ithaca, NY 14853

ABSTRACT

Adequate supply of high-quality colostrum is essential for calf health. Colostrum production, at first milking, varies between animals and seasons, but herd-level and management associations with colostrum production have not been well described. Our objectives were to (1) describe colostrum production and colostrum handling practices and (2) to identify individual cow, herd management, and environmental factors associated with colostrum production. A convenience sample of 19 New York Holstein dairy farms (620 to 4,600 cows) were enrolled in this observational study to describe colostrum production and to evaluate cow, management, and prepartum environmental factors associated with colostrum yield and Brix %. Herd owners or managers were given a colostrum management questionnaire, and farm personnel recorded individual colostrum yield and Brix % for primiparous (PP; n = 5,978) and multiparous (MPS; n = 13,228) cows between October 2019 and February 2021. Temperature, relative humidity, and light intensity were measured by sensors placed in each farm's close-up dry cow pens for the entire length of the study. Median colostrum yield for each farm ranged from 2.5 to 7.6 kg for PP and 4.0 to 7.7 kg for MPS cows. Mean Brix % from each farm ranged from 22.2 to 27.9% for PP and 22.0 to 28.8% for MPS cows. Lowest colostrum yield from PP animals was associated with calf sex (female) and colostrum Brix % (≤22%). Greatest colostrum yield from MPS cows was associated with colostrum Brix % (≤22%), calf sex (twin), dry period length (>67 d), gestation length (283–293 d), an alive calf, second parity, previous lactation length (>344 d) and previous lactation 305-d mature equivalent milk yield (>13,091 kg), heat and humidity exposure area under the curve (AUC) 7 d before calving (>69.2 average temperature-humidity index per 30-min interval), and light intensity AUC 14 d before calving (>154.2 average lux per 15-min interval). Greatest colostrum Brix % from PP animals was associated with calf sex (male), an alive calf, and light intensity AUC 14 d before calving (≤64.0 average lux per 15-min interval). Greatest colostrum Brix % from MPS cows was associated with dry period length (>67 d), an alive calf, 305-d mature equivalent milk yield of previous lactation (≤15,862 kg), gestation length (274–282 d), colostrum yield (<6 kg), fifth or greater parity, and heat and humidity exposure AUC 7 d before calving (≤50.1 average temperature-humidity index per 30-min interval). Dairy producers can use this information to recognize the variation in colostrum production and alter colostrum management programs in anticipation of periods of low production or quality.
Key words: colostrum, Brix %, epidemiology, temperature-humidity index, management

INTRODUCTION

Newborn calves rely on colostrum for essential nutrients, hormones, growth factors, and other components for early life nutrition, organ and intestinal development, and transfer of passive immunity (Fischer-Tlustos et al., 2021; Lopez and Heinrichs, 2022); thus, maintaining an adequate supply of high-quality colostrum is essential for raising healthy calves. Colostrum quality and yield have been shown to vary by cow, month of calving, and season (Conneely et al., 2013; Gavin et al., 2018; Borchardt et al., 2022); however, the mechanisms regulating colostrum production in dairy cattle remain poorly understood. Cows entering parity ≥3 are reported to have higher immunoglobulin G (**IgG**) concentration compared with cows in parity 1 or 2 (Bartier et al., 2015). Shortening the dry period does not appear to affect colostrum quality, but reduction in colostrum yield has been reported (Mansfeld et al., 2012; Mayasari et al., 2015; O'Hara et al., 2019). Maximum temperature-humidity index (**THI**) and photoperiod (Gavin et al., 2018) as well as season (Conneely et al., 2013; Borchardt et al., 2022) have been associ-

Received June 25, 2022.
Accepted September 12, 2022.
*Corresponding author: sm682@cornell.edu

ated with changes in colostrum production and quality. Other studies, however, have shown colostrum quality not to be associated with season or month of calving (Pritchett et al., 1991; Bartier et al., 2015; Dunn et al., 2017). In addition, experimental manipulation of the photoperiod during the dry period did not affect colostrum yield or IgG concentration (Morin et al., 2010). There is a lack of data available to understand factors that are associated with colostrum production in multiple herds and in year-round calving confined management systems.

The gold standard for assessing colostrum quality is the quantification of IgG concentration by radial immunodiffusion (Ahmann et al., 2021). Although this is an accurate method, the cost and time associated with submitting colostrum samples to the laboratory for this analysis makes it infeasible for farms (Godden, 2008). Refractive index measurement, for example, using a Brix refractometer, is a rapid method for estimation of colostral IgG concentration, and has shown moderate to strong correlation (r = 0.64 to 0.75) with radial immunodiffusion (Bielmann et al., 2010; Quigley et al., 2013; Bartier et al., 2015), thus providing an acceptable tool for on farm use.

We hypothesized that colostrum yield and Brix % are associated with cow, farm management, and environmental factors. Our objectives of this work performed on New York Holstein dairy farms were to (1) describe colostrum production and colostrum handling practices and (2) identify individual cow, herd management, and environmental factors associated with colostrum production.

MATERIALS AND METHODS

Farm Selection and Enrollment

All procedures were approved by the Cornell University Institutional Animal Care and Use Committee (protocol number 2019–0058). A list of farm contacts was compiled by the investigators based on previous New York statewide research projects, as well as by contacting New York veterinarians and nutritionists. The following were inclusion criteria: (1) ability to collect and record individual colostrum weight or volume and a composite sample Brix % reading, (2) minimum herd size of 500 lactating Holstein cows, (3) use of dairy management software DairyComp 305 (DC305, Valley Ag Software), and (4) heifer calves housed on site for at least the first week of life. A total number of 51 farms were contacted from October 2019 to January 2020. Owners or herd managers were asked, by phone or email, if the farm met the inclusion criteria and if they were interested in participating in the study. Farms not enrolled did not meet inclusion criteria (n = 4), were not willing to participate (n = 12), or failed to respond (n = 16). At the end of the enrollment period, a convenience sample of 19 New York Holstein dairy farms were included in this observational study between October 2019 and February 2021.

Farm Survey

A written consent form and a survey of transition cow, calf, and colostrum management were completed with the farm owner or herd manager during the initial enrollment farm visit. The survey was comprised of both open and closed questions related to dry off procedures, pen movement, dry period nutrition strategies, transition cow housing, calving environment, colostrum harvest protocols, colostrum management and feeding, and preweaning calf management (Supplemental File 1; https://hdl.handle.net/1813/112245; Westhoff et al., 2022).

Colostrum Records

Farm personnel harvested individual cow colostrum according to existing farm protocols. Colostrum yield was either collected as a volume or as a weight. Weight was measured on a digital scale and volume was determined using volume markers on commercial bottles or milking buckets. For farms choosing to collect colostrum yield in weight (n = 15), colostrum collection buckets were labeled with a unique colored band, and empty weights were recorded for each bucket. Farm personnel were then instructed to record the bucket color and the total weight of the colostrum bucket such that the colostrum yield could be calculated. A Brix refractometer (Model PA201, Misco) was provided to each farm and farm personnel were trained on the correct use of the instrument for composite sample Brix % reading. This included providing and reviewing a standard operating procedure for reading colostrum Brix % and cleaning the refractometer. Colostrum record binders were provided to each farm to record cow ID, date and time of colostrum harvest, bucket color, colostrum yield, Brix %, notes, and the initials of the individual responsible for colostrum collection. Final colostrum yield was calculated by subtracting the recorded weight of the bucket from the total weight. For farms collecting colostrum yield as volume (n = 4), farm personnel recorded volume in pints or liters. Colostrum volume was converted to liters and then to weight using the equations $L = P \times 2.1134$ and kg = $L \times 1.0524$, where L = colostrum volume in liters, P

= colostrum volume in pints, kg = colostrum weight in kilograms, and 1.0524 is the density of Holstein colostrum (Morin et al., 2001).

Environmental Data

Two environmental data loggers measuring light intensity (lum/ft^2) and ambient temperature/relative humidity (HOBO Models MX2202/MX2301A, respectively, Onset Computer Corp.) were mounted facing the length of the barn, approximately 3 m above ground, directly above the resting area in the close-up dry cow pen (defined as the pen housing animals for a minimum of 3 wk before calving) at each farm. Light intensity and temperature/relative humidity were recorded in 15- and 30-min intervals during the entire study period, respectively. Temperature-humidity index was calculated using the equation

$$THI = T - \left\{0.55 \times \left[0.55 \times \left(\frac{RH}{100}\right)\right]\right\} \times \left(T - 58\right),$$

where T = ambient temperature (°F) and RH = relative humidity percent (Gavin et al., 2018). Lux was calculated using the equation Lux = $LUM \times 10.764$, where LUM = lumen/ft^2.

Farm Visits and Data Collection

Farms were visited 4 times, approximately 3 mo apart, during the data collection period. At each visit, colostrum records and a DC305 backup was collected. Refractometers were cleaned, zero-set with distilled water, and calibrated (refractometer calibration fluid, Misco) according to the manufacturer's instructions.

Analytical Approach

A formal sample size for the inclusion of the number of farms was not calculated for this observational study. With the objective to describe the epidemiology of colostrum production across New York state, we aimed to enroll as many herds as possible that met the inclusion criteria and agreed to participate during our enrollment period of October 2019 to January 2020.

Reports were generated from herd records that included cow ID, calving date, parity, breed, days dry, gestation length, previous lactation length, age at first calving, 305-d mature equivalent milk yield (**305ME**) of previous lactation, sex of the calf, whether the calf was a stillbirth (defined as DC305 code "dead on arrival"), days on the close-up diet (**DONCU**), and singleton or twin. Colostrum records were entered into Microsoft Excel (version 2202, Microsoft Corp.) and merged with DC305 reports into a single data set using cow ID, date of calving, and colostrum harvest.

For primiparous (**PP**) cows, independent variables considered for associations with colostrum yield and Brix % included sex of the calf, age at first calving, whether the calf was a stillbirth, colostrum yield and Brix %, gestation length, heat and humidity exposure, and light intensity. For multiparous (**MPS**) cows, variables included in univariable screening for associations with colostrum yield and Brix % included sex of the calf, whether the calf was a stillbirth, parity, colostrum yield and Brix %, gestation length, days dry, heat and humidity exposure, light intensity, and previous lactation length and 305ME. Days on the close-up diet was considered for associations with colostrum yield and Brix % on a subset of records from PP and MPS cows.

Continuous variables (age at first calving, Brix %, colostrum yield, gestation length, heat and humidity exposure, light intensity, dry period length, DONCU, and previous lactation length and 305ME) were first assessed for a linear relationship with colostrum yield and Brix %, respectively. If the assumption of a linear relationship was not fulfilled (defined as an absolute correlation coefficient ≥0.20), variables were categorized for subsequent analysis. Individual cow records with a recorded gestation length greater or less than 15 d of the mean were removed to limit inclusion of animals with incorrect records of breeding dates or abortions (Norman et al., 2009). Gestation length was categorized for both PP and MPS into the following 3 categories: short (PP = 261–271, MPS = 263–273 d), normal (PP = 272–280, MPS = 274–282 d), or long (PP = 281–291, MPS = 283–293 d). Brix % was grouped into the following 4 categories: ≤22.0, 22.1 to 24.4, 24.5 to 27.0, and >27.0%. Colostrum yield was dichotomized at <6 and ≥6 kg as the amount of colostrum needed for 2 colostrum feedings (3.78 and 1.89 L at first and second feeding). Age at first calving, dry period length, and DONCU were grouped into 3 categories as follows: ≤20, 21–24, >24 mo; <47, 47–67, >67 d; and ≤15, 16–30, >30 d, respectively. Quartiles 1 and 3 were used as cut point for 3 categories of previous lactation length (<297, 297–344, >344 d) and previous lactation 305ME (≤13,090, 13091–15,862, >15,862 kg). Due to fewer animals entering parities 6 to 10, they were grouped together resulting in parity categories 1, 2, 3, 4, or ≥5 (5+). Given the low number of twin calvings, twins were not further categorized by sex, resulting in the 3 calf categories singleton female, singleton male, or twins.

To account for the total exposure to light intensity (lux), as well as heat and humidity exposure (THI) during the close-up period, total area under the curve (**AUC**) was calculated separately for the following 5 different periods in the last 3 wk of the prepartum period: −21 to −1, −14 to −1, −21 to −15, −14 to −8, and −7 to −1 d, relative to calving using the trapezoidal method (trapz function) in a data analysis software (MatLab R2022a, The MathWorks Inc.). Light intensity AUC was categorized into 3 groups using quartiles 1 and 3 as cut points for each prepartum period. Heat and humidity exposure (THI) AUC was categorized for each prepartum period for an average THI per 30-m interval of ≤40.2, 40.3 to 50.1, 50.2 to 60.0, 60.1 to 69.2, and >69.2. To prevent multicollinearity, each prepartum period was screened in univariable models with dependent variables colostrum yield, 0 kg recorded colostrum yield, and Brix %. The prepartum period with the lowest Akaike information criterion was then selected for further analysis.

Following univariable screening, separate mixed effects multivariable models were generated in PROC MIXED (SAS 9.4, SAS Institute Inc.) for the following 4 outcomes of interest: colostrum yield from PP cows, Brix % from PP cows, colostrum yield from MPS cows, and Brix % from MPS cows. All mixed models included the random effects of herd and month of calving. All variables with $P \leq 0.10$ in univariable screening were included in the initial multivariable model. Stepwise manual backward elimination was used until a final model was defined as all remaining variables having $P < 0.05$. Biologically plausible 2-way interactions were then tested and retained in the model if $P < 0.05$. The assumptions of normality and homoscedasticity of the residuals were visually assessed, and colostrum yield was transformed using the natural logarithm to meet the model assumptions. Tukey's post hoc test was used to adjust pairwise comparisons for the number of multiple comparisons.

Mixed effects multivariable models accounting for herd and month of calving were generated using PROC MIXED (SAS v. 9.4) on a subset of records with available DONCU for the outcome variables of colostrum yield from PP cows, Brix % from PP cows, colostrum yield from MPS cows, and Brix % from MPS cows. Variables with $P \leq 0.10$ in univariable screening entered the initial multivariable model. Stepwise manual backward elimination was used until all remaining variables had $P \leq 0.05$. The assumptions of normality and homoscedasticity of the residuals were visually assessed for all models, and data for colostrum yield were transformed using the natural logarithm to meet the model assumptions.

In a logistic regression analysis, variables associated with the binary outcome of 0 kg or >0 kg of recorded colostrum yield were separately investigated for PP and MPS cows accounting for herd as a random effect using PROC GLIMMIX (SAS v. 9.4). All variables were first screened in a univariable model and entered the initial logistic regression model when $P \leq 0.20$. Stepwise manual backward elimination was used until all remaining variables had $P \leq 0.10$.

RESULTS

Enrolled Herds

Nineteen herds with average (range) herd size of 1,545 (620 to 4,600) cows were enrolled in the study. Staff at farm T was not able to collect Brix % and colostrum yield; therefore, the cows from this herd were not included in the analysis. Farms N and U were managed together where farm N housed PP and MPS animals and farm U housed MPS animals. Due to a Cornell University mandated COVID-19 shutdown of all research activities from March to June 2020, a quarterly visit could not be completed for all farms such that 6 farms (31.6%) were visited 4 times and 13 farms (68.4%) were visited 3 times during the data collection period.

During the study period 21,374 individual animal colostrum records were collected from 18 herds. After removing records for breeds other than Holstein (n = 250), missing cow ID (n = 372), and missing colostrum yield (n = 139), 1,016 records had a 0-kg recorded colostrum yield and 19,597 records had >0-kg recorded colostrum yield (Figure 1). Additional records were removed from the initial data set before data analysis for the following reasons: gestation length ± 15 d of the mean (n = 246), or an incomplete data set (n = 1,101). After removal of these data, analysis was performed on 18,343 and 923 records with >0- and 0-kg recorded colostrum yield, respectively. An analysis of a subset of data with available DONCU was performed on 4,766 and 11,266 records from PP and MPS animals, respectively. Descriptive statistics for colostrum yield and Brix % for records with >0-kg recorded colostrum yield and the percentage of animals with a final colostrum record out of the total number of calvings for each herd during the data collection period are presented in Table 1. Median and range daily THI and light intensity by month are presented in Figure 2.

Colostrum Management

Preweaning calf and colostrum management practices are presented in Table 2. On most farms, colostrum was

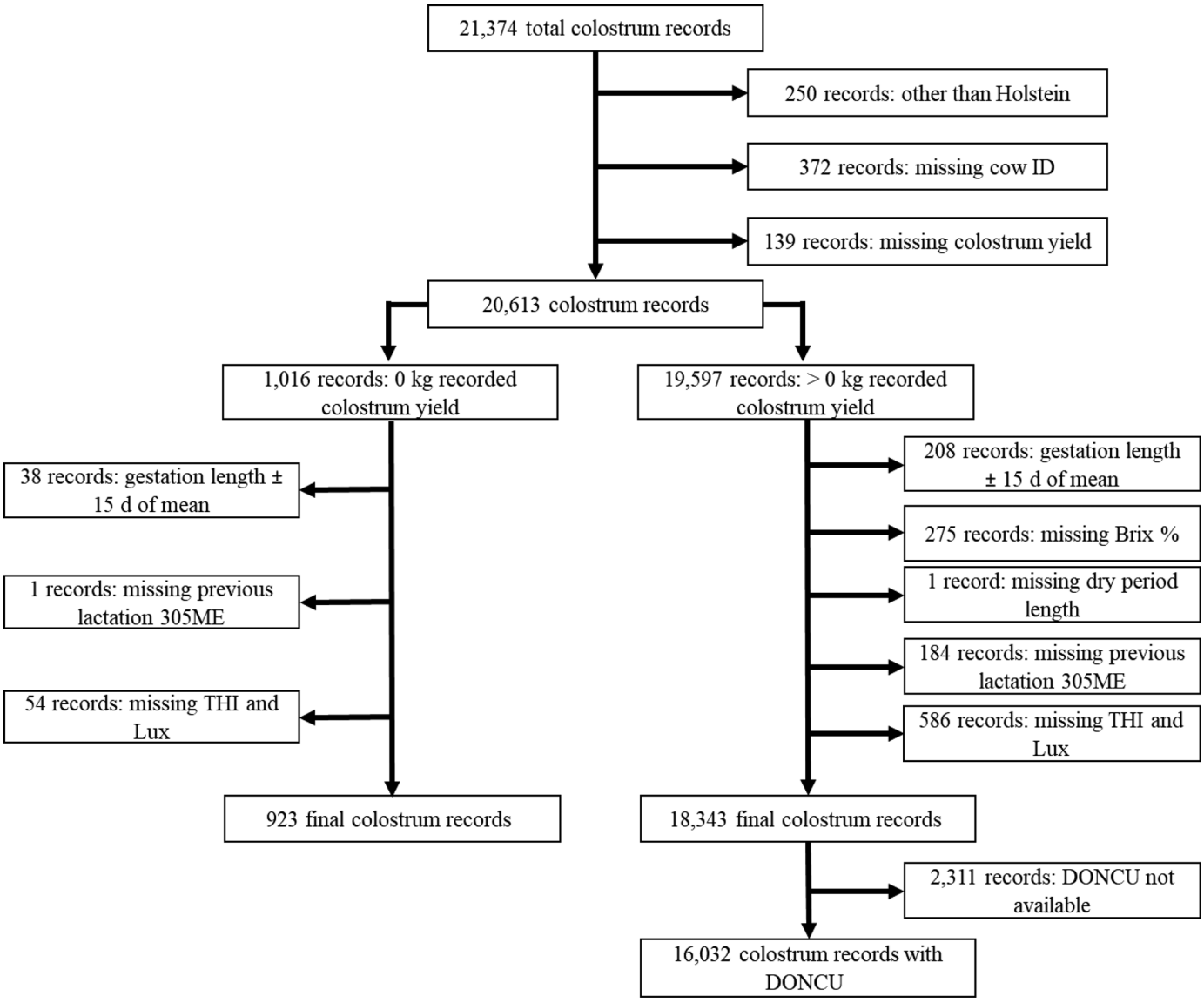


Figure 1. Flowchart of colostrum records included in statistical analysis. Records were collected from a convenience sample of 18 New York Holstein dairy farms. 305ME = 305-d mature equivalent milk yield; THI = temperature-humidity index; DONCU = days on the close-up diet.

collected in the parlor or hospital parlor within 8 to 15 h of calving. Common colostrum handling practices at time of enrollment included refrigeration, reading colostrum Brix %, and selling colostrum for commercial use. On a subset of farms, colostrum was discarded for visual abnormality (n = 9, 47.4%), oversupply (n = 2, 10.5%), or not meeting farm specific minimum Brix % (n = 2, 10.5%). On almost all farms, a first colostrum volume of 3.78 L was fed to heifer and bull calves (Table 2).

Colostrum Yield

Descriptive statistics for cows with a >0-kg recorded colostrum yield is presented in Tables 3 and 4. Median colostrum yield was highest in parity 2 and lowest in parity 1. Box and whisker plots for monthly colostrum yield from PP and MPS cows are presented in Figures 3A and 3B, respectively. Monthly colostrum yield by farm is presented in Figure 4.

Colostrum Yield from Primiparous Cows

After univariable screening, the variables offered to the multivariable mixed model included sex of the calf, Brix % (categorized), gestation length (categorized), age at first calving (categorized), heat and humidity exposure AUC 7 d before calving (categorized), and light intensity AUC 14 d before calving (categorized).

Table 1. Descriptive statistics of colostrum production from a convenience sample of 19 New York Holstein dairy farms

					Primiparous[2]		Multiparous	
Farm	Herd size	Colostrum record collection (mo/yr)	% of calvings recorded[1]	n	Median (Q1, Q3) yield	Mean (SD) Brix %	Median (Q1, Q3) yield	Mean (SD) Brix %
A	620	10/2019 to 1/2021	79.8	731	4.2 (2.6, 6.2)	24.3 (3.4)	6.3 (4.1, 9.5)	26.1 (4.4)
B	1,535	10/2019 to 10/2020	32.2	548	4.2 (2.9, 8.4)	22.2 (2.6)	4.2 (2.1, 8.4)	22.0 (2.7)
C	1,075	11/2019 to 1/2021	91.0	1,380	5.9 (4.5, 8.2)	25.5 (4.0)	5.9 (3.6, 9.1)	26.5 (4.6)
D	1,100	12/2019 to 6/2020	20.3	139	7.6 (5.8, 8.2)	25.1 (3.9)	7.7 (6.2, 8.9)	27.6 (3.7)
E	1,310	10/2019 to 1/2021	71.7	1,480	3.0 (2.0, 5.0)	23.3 (3.5)	4.0 (2.0, 6.0)	24.4 (4.1)
F	1,220	11/2019 to 7/2020	11.3	97	6.0 (3.5, 8.3)	25.7 (3.2)	7.0 (5.0, 11.1)	26.2 (4.7)
G	2,100	1/2020 to 1/2021	79.1	2,057	3.0 (2.0, 4.0)	26.0 (3.2)	4.0 (2.0, 6.0)	26.3 (3.7)
H	1,325	11/2019 to 1/2021	68.5	1,309	5.6 (3.5, 8.0)	22.9 (3.3)	6.3 (3.5, 9.2)	24.9 (3.8)
I	680	12/2019 to 1/2021	91.1	756	4.7 (2.6, 7.3)	23.3 (3.3)	6.6 (3.7, 9.7)	25.6 (4.1)
K	1,830	12/2019 to 1/2021	89.9	2,120	4.5 (3.4, 6.4)	23.3 (3.7)	4.5 (2.3, 6.8)	26.6 (4.8)
L	1,550	1/2020 to 1/2021	98.1	1,815	3.5 (2.1, 4.8)	26.3 (3.4)	4.3 (2.5, 6.1)	26.9 (4.2)
M	1,800	1/2020 to 1/2021	44.2	911	3.0 (2.0, 5.1)	24.1 (3.3)	4.0 (2.0, 6.0)	25.5 (4.0)
N	1,435	1/2020 to 2/2021	19.5	330	3.6 (1.4, 6.4)	24.3 (3.2)	5.4 (3.4, 7.6)	25.3 (2.5)
O	720	12/2019 to 1/2021	84.3	720	3.9 (2.4, 5.9)	27.5 (4.9)	4.8 (2.8, 7.0)	28.8 (5.1)
P	1,090	1/2020 to 1/2021	72.2	759	3.3 (2.4, 4.1)	27.9 (4.1)	4.2 (2.9, 6.0)	26.9 (4.6)
R	1,000	1/2020 to 1/2021	22.8	249	2.5 (1.6, 3.5)	24.2 (4.9)	5.1 (3.2, 7.8)	26.4 (5.7)
S	2,450	2/2020 to 1/2021	69.6	2,001	5.5 (3.4, 7.8)	23.1 (3.9)	5.3 (2.9, 8.3)	23.6 (3.7)
T[3]	4,600	N/A[4]	N/A	N/A	N/A	N/A	N/A	N/A
U[5]	1,920	1/2020 to 1/2021	75.3	1,864	N/A	N/A	5.7 (2.7, 8.8)	25.4 (4.3)
Total	N/A	N/A	65.5	19,266	4.0 (2.3, 6.2)	24.6 (3.9)	4.6 (2.7, 7.7)	25.7 (4.4)

[1]Calculated as the number of colostrum records included in statistical analysis divided by the number of calvings during colostrum record collection.

[2]Q = quarter.

[3]Farm was unable to collect colostrum yield and Brix %.

[4]N/A = not applicable.

[5]Farm housed 100% multiparous cows.

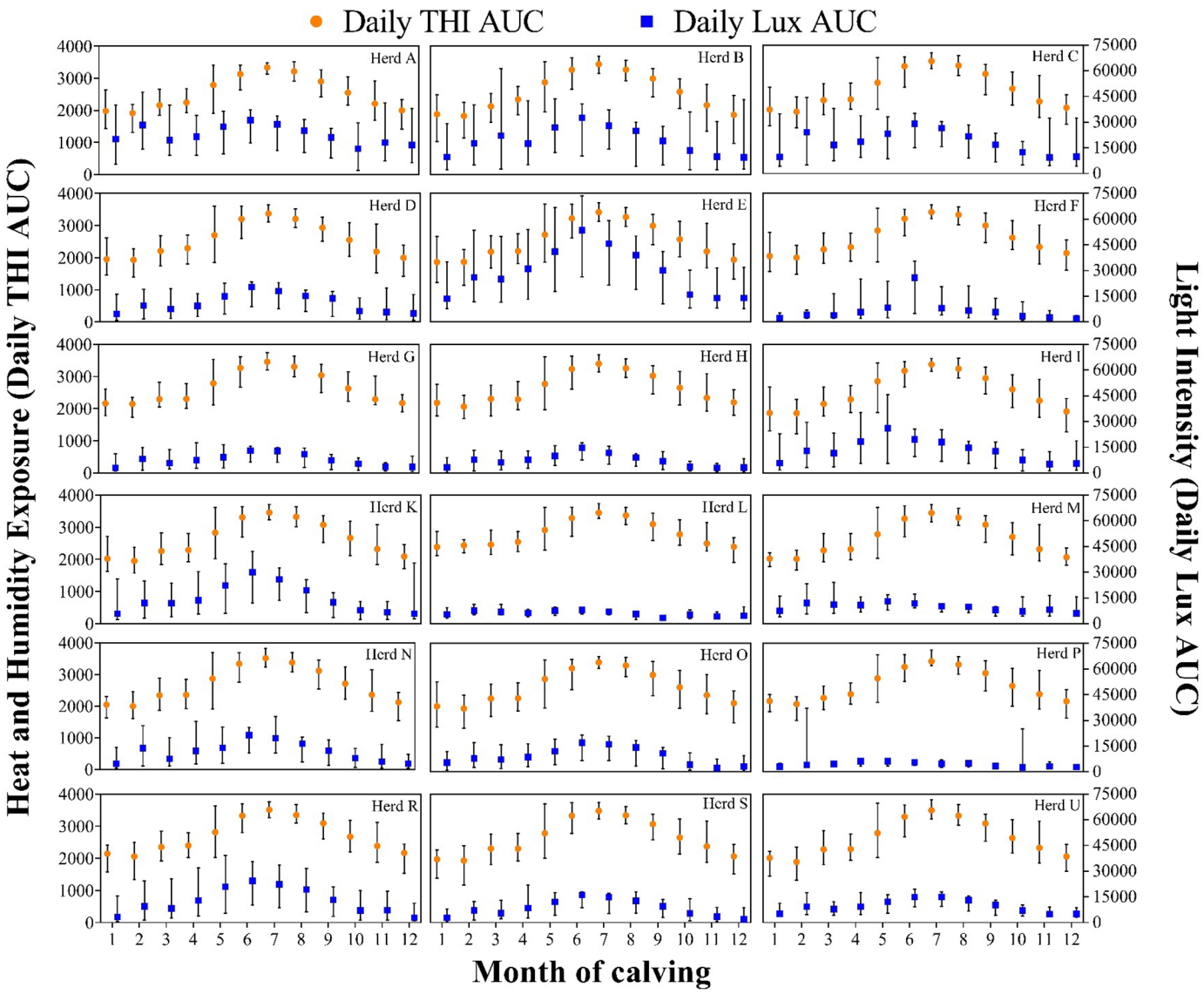


Figure 2. Monthly median and range daily temperature-humidity index (THI) area under the curve (AUC; yellow) and daily light intensity (lux) AUC (blue) from a convenience sample of 18 New York Holstein dairy farms. We collected THI and lux in 30- and 15-min intervals, respectively, approximately 3 m above ground directly above the resting area in the close-up dry cow pen at each enrolled farm. Each panel contains the respective herd code.

In the final model (Table 5), colostrum yield from PP cows was associated with sex of the calf ($P = 0.008$) and categorized colostrum Brix % ($P < 0.001$) when controlling for herd ($P = 0.004$) and month of calving ($P = 0.018$). Colostrum yield was highest with a male calf and in Brix categories 22.1 to 24.4% and 24.5 to 27.0%, and lowest when Brix $\leq$ 22%.

Colostrum Yield from Multiparous Cows

After univariable screening, the variables offered to the multivariable mixed model included sex of the calf, whether the calf was a stillbirth, parity, Brix % (categorized), dry period length (categorized), previous lactation 305ME (categorized), gestation length (categorized), previous lactation length (categorized), heat and humidity exposure AUC 7 d before calving (categorized), and light intensity AUC 14 d before calving (categorized). The final multivariable model is shown in Table 5. Controlling for month of calving ($P = 0.017$) and herd ($P = 0.003$), colostrum yield from MPS cows was associated with sex of the calf ($P < 0.001$), stillbirth ($P = 0.006$), parity ($P < 0.001$), categorized colostrum Brix % ($P < 0.001$), dry pe-

Table 2. Preweaned calf and colostrum management practices used by a convenience sample of 19 New York Holstein dairy farms

Variable	Farm (n)	%[1]
Location of colostrum harvest		
Maternity pen	4	21.1
Hospital parlor	2	10.5
Parlor	13	68.4
Range of time to colostrum harvest (h)		
0–2	3	15.8
0–8	10	52.6
0–15	5	26.3
0–24	1	5.3
Oxytocin use		
Yes	11	57.9
No	8	42.1
Colostrum handling practices[2]		
Feed to dam's calf	2	10.5
Freeze	8	42.1
Refrigerate	13	68.4
Heat treat	3	15.8
Pool	7	36.8
Read colostrum Brix %	14	73.7
Colostrum sold	13	68.4
Discard	11	57.9
Range of time to first colostrum feeding (h)		
0–2	18	94.7
0–12	1	5.3
First colostrum feeding method		
Esophageal feeder	13	68.4
Bottle	2	10.5
Bottle and esophageal feeder	4	21.1
Volume of first colostrum feeding to heifer calves (L)		
2.84	1	5.3
3.78	18	94.7
Number of colostrum feedings to heifer calves		
1	5	26.3
2	11	57.9
3	3	15.8
Volume of first colostrum feeding to bull calves (L)		
1.89	2	10.5
2.84	1	5.3
3.78	16	84.2
Number of colostrum feedings to bull calves		
1	12	63.2
2	7	36.8
Preweaned calf housing[2]		
Hutch	9	47.4
Individual pen	5	26.3
Group	6	31.6
Milk fed to preweaned calves		
Milk replacer	4	21.1
Acidified waste milk	2	10.5
Pasteurized waste milk	7	36.8
Raw waste milk	2	10.5
Pasteurized whole milk	2	10.5
Raw whole milk	2	10.5

[1]Calculated as number of farms divided by 19.

[2]Farms could use multiple management practices.

riod length ($P < 0.001$), previous lactation 305ME ($P < 0.001$), gestation length ($P < 0.001$), previous lactation length ($P < 0.001$), heat and humidity exposure AUC 7 d before calving ($P < 0.001$), and light intensity AUC 14 d before calving ($P = 0.012$). Colostrum yield increased with increasing dry period length and gestation length categories. A stillbirth, female calf, previous lactation 305ME less than 13,091 kg, and previous lactation length less than 345 d were associated with reduced colostrum yield. Increasing colostrum Brix % categories were associated with decreasing colostrum yield. Greatest colostrum yield was associated with

Table 3. Descriptive statistics for 18,343 Holstein cows from a convenience sample of 18 New York dairy farms

Variable	Parity	Mean	Median	SD	Range
Colostrum yield (kg)	1	4.72	4.08	2.92	0.1–38.6
	2	6.05	5.45	3.78	0.1–29.6
	3	5.87	4.97	3.93	0.1–43.8
	4	5.60	4.54	3.97	0.1–33.1
	5+	5.66	4.54	4.07	0.1–41.8
Brix %	1	24.61	24.5	3.92	10.3–47.2
	2	24.61	24.3	4.03	10.9–50.9
	3	25.84	25.6	4.38	7.0–58
	4	26.71	26.5	4.29	8.4–47.3
	5+	27.61	27.4	4.78	9.7–50.7
Gestation length (d)	1	275.81	276	4.66	261–291
	2	277.24	277	4.76	263–293
	3	278.18	278	4.90	263–293
	4	278.72	279	4.82	263–293
	5+	278.78	279	5.09	263–293
Age at first calving (mo)	1	21.90	22	1.46	17–39
Dry period length (d)	2	56.34	56	9.21	25–156
	3	57.25	55	11.18	20–134
	4	58.65	56	12.54	23–161
	5+	58.28	56	11.89	33–131
Previous lactation length (d)	2	323.92	307	45.35	224–624
	3	324.41	308	44.08	204–571
	4	324.63	307	46.90	220–652
	5+	326.85	311	45.60	246–570
Previous lactation 305ME[1] (kg)	2	14,577	14,576	2,214	5,416–26,758
	3	14,819	14,801	2,166	5,543–29,493
	4	14,260	14,201	2,080	6,819–26,542
	5+	13,796	13,748	1,921	7,613–26,324
Heat and humidity exposure AUC[2] (THI 7 d before calving)	1	18,400	17,361	3,665	9,650–25,697
	2	18,364	17,354	3,392	9,655–25,697
	3	18,411	17,772	3,751	9,650–25,697
	4	18,470	18,075	3,715	9,650–25,697
	5+	18,427	18,046	3,783	9,650–25,697
Light intensity AUC[3] (lux 14 d before calving)	1	164,796	129,075	1,227	21,989–837,874
	2	167,338	132,721	122,286	23,007–837,874
	3	170,414	137,566	121,198	21,989–826,387
	4	180,870	148,828	125,374	21,989–837,874
	5+	181,673	149,211	120,850	23,670–778,635
Days on close-up diet[4] (d)	1	25.98	25	12.16	1–132
	2	33.11	28	16.98	1–128
	3	34.76	30	16.49	1–118
	4	37.47	32	17.68	2–120
	5+	37.74	33	16.63	1–113

[1]305ME = 305-d mature equivalent milk yield.

[2]Area under the curve (AUC) was calculated for temperature-humidity index (THI) 7 d before calving. THI was collected from the close-up dry cow pen every 30 min.

[3]Area under the curve was calculated for light intensity (lux) 14 d before calving. Lux was collected from the close-up dry cow pen every 15 min.

[4]Days on the close-up diet was investigated on a subset of 4,766 primiparous and 11,266 multiparous cows.

parity 2, average THI per 30-min interval 7 d before calving >69.2, and average lux per 15-min interval 14 d before calving >154.2.

Colostrum Brix %

Descriptive statistics for colostrum Brix % by parity are presented in Table 3. A subset of PP (n = 1,347, 22.5%) and MPS (n = 2,315, 18.4%) cows produced colostrum below the industry standard cut point, for high-quality colostrum (22% Brix). Box and whisker plots of PP and MPS monthly colostrum Brix % are presented in Figures 3C and 3D, respectively.

Colostrum Brix % from Primiparous Cows

Colostrum yield (categorized), gestation length (categorized), sex of the calf, whether the calf was a stillbirth, heat and humidity exposure AUC 7 d before calving (categorized), and light intensity AUC 14 d before calving (categorized) were included as fixed effects in the initial multivariable mixed model after univariable screening. In the final multivariable model, calf sex ($P < 0.001$), stillbirth ($P = 0.013$), and light intensity AUC 14 d before calving ($P = 0.011$) controlling for colostrum yield ($P = 0.095$), month of calving ($P = 0.019$), and herd ($P = 0.003$) remained

Table 4. Descriptive statistics of categorical variables for 5,790 primiparous and 12,553 multiparous Holstein cows from 18 New York dairy farms

Variable	Primiparous (%)	Multiparous (%)
Colostrum yield (kg)		
<6	73.4	61.5
≥6	26.6	38.5
Brix %		
≤22	23.9	20.4
22.1–24.4	25.7	20.8
24.5–27	26.1	24.0
>27	24.3	34.8
Calf sex		
Female	66.0	47.3
Male	33.5	48.2
Twin	0.5	4.5
Stillbirth		
Alive	96.3	97.8
Dead	3.7	2.2
Heat and humidity exposure		
AUC 7 d before calving[1] (average THI per 30-min interval)		
≤40.2	5.5	6.3
40.3–50.1	39.7	38.8
50.2–60.0	19.8	19.3
60.1–69.2	20.3	20.9
>69.2	14.7	14.7
Light intensity AUC		
14 d before calving[2] (average lux per 15-min interval)		
≤64.0	26.8	23.5
64.1–154.2	49.8	51.2
>154.2	23.4	25.3
Gestation length[3] (d)		
Short	16.6	18.3
Normal	67.4	63.8
Long	16.00	17.9
Age at first calving (mo)		
≤20	6.3	
21–24	88.0	
>24	5.7	
Dry period length (d)		
<47		9.8
47–67		79.6
>67		10.6
Previous lactation length (d)		
<297		26.9
297–344		48.4
>344		24.7
Previous lactation 305ME[4] (kg)		
≤13,090		24.9
13,091–15,862		50.2
>15,862		24.9
Parity		
2		40.9
3		29.7
4		16.7
5+		12.7
Days on close-up diet[5] (d)		
≤15	15.1	5.3
16–30	55.8	47.2
>30	29.1	47.5

[1]Area under the curve (AUC) was calculated for temperature-humidity index (THI) 7 d before calving in 30-min intervals. THI was collected from the close-up dry cow pen at each farm.

[2]Area under the curve was calculated for light intensity (lux) 14 d before calving in 15-min intervals. Lux was collected from the close-up dry cow pen at each farm.

[3]Primiparous (PP) and multiparous (MPS) gestation length: short (PP = 261–271, MPS = 263–273 d), normal (PP = 272–280, MPS = 274–282 d), or long (PP = 281–291, MPS = 283–293 d).

[4]305ME = 305-d mature equivalent milk yield.

[5]Days on the close-up diet was investigated on a subset of 4,766 primiparous and 11,266 multiparous cows.

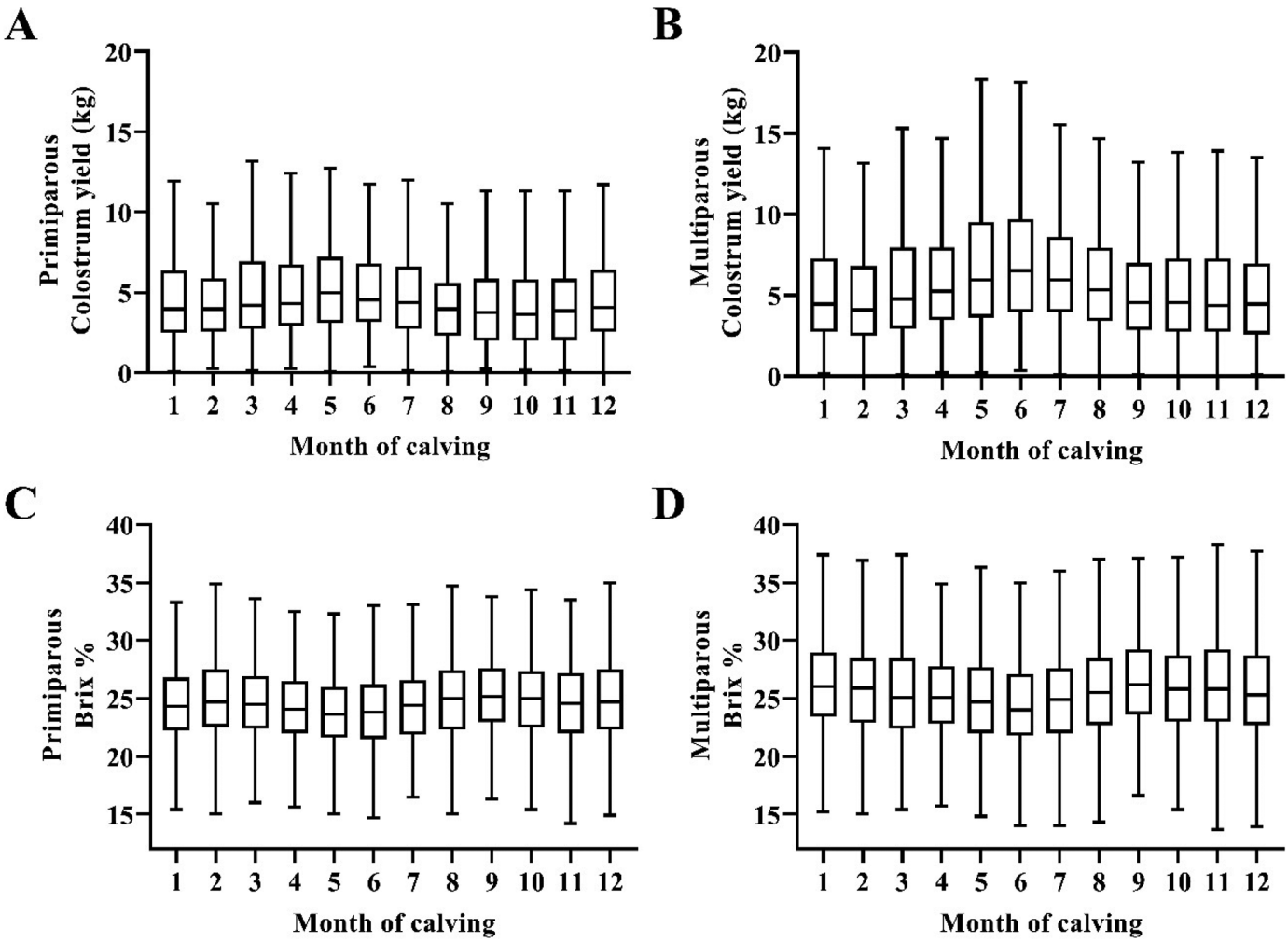


Figure 3. Box and whisker plots of monthly colostrum yield (kg) and Brix % from 5,790 primiparous and 12,553 multiparous Holstein cows from 18 New York farms. (A) Monthly colostrum yield from primiparous cows; (B) monthly colostrum yield from multiparous cows; (C) monthly colostrum Brix % from primiparous cows; (D) monthly colostrum Brix % from multiparous cows. Boxes = first and third quartiles; mid-line = median; lower whiskers = quartile 1 − (interquartile range × 1.5); upper whisker = quartile 3 + (interquartile range × 1.5).

in the model (Table 6). Colostrum Brix % was lowest with a stillbirth, average lux per 15-min interval 64.1 to 154.2, and female calf.

Colostrum Brix % from Multiparous Cows

The initial model after univariable screening contained colostrum yield (categorized), gestation length (categorized), dry period length (categorized), previous lactation 305ME (categorized), calf sex, whether the calf was a stillbirth, parity, previous lactation length (categorized), heat and humidity exposure AUC 7 d before calving (categorized), and light intensity AUC 14 d before calving (categorized) as fixed effects. Variables remaining in the final multivariable model (Table 6) for Brix % of colostrum from MPS cows when accounting for month of calving ($P = 0.018$) and herd ($P = 0.002$) were stillbirth ($P = 0.005$), parity ($P < 0.001$), heat and humidity exposure AUC 7 d before calving ($P = 0.003$), dry period length ($P < 0.001$), previous lactation 305ME ($P = 0.005$), gestation length ($P = 0.01$), and colostrum yield ($P < 0.001$). Brix % was greater for cows with a dry period >67 d compared with <47 or 47 to 67 d. Increasing parity was associated with increased colostrum Brix %. Colostrum Brix % was greatest with an alive calf, previous lactation 305ME ≤15,862 kg, and average THI per 30-min interval ≤50.1.

Primiparous Animals With Known Days on Close-up Diet

Descriptive statistics for days on close-up diet are presented in Tables 3 and 4. After univariable screen-

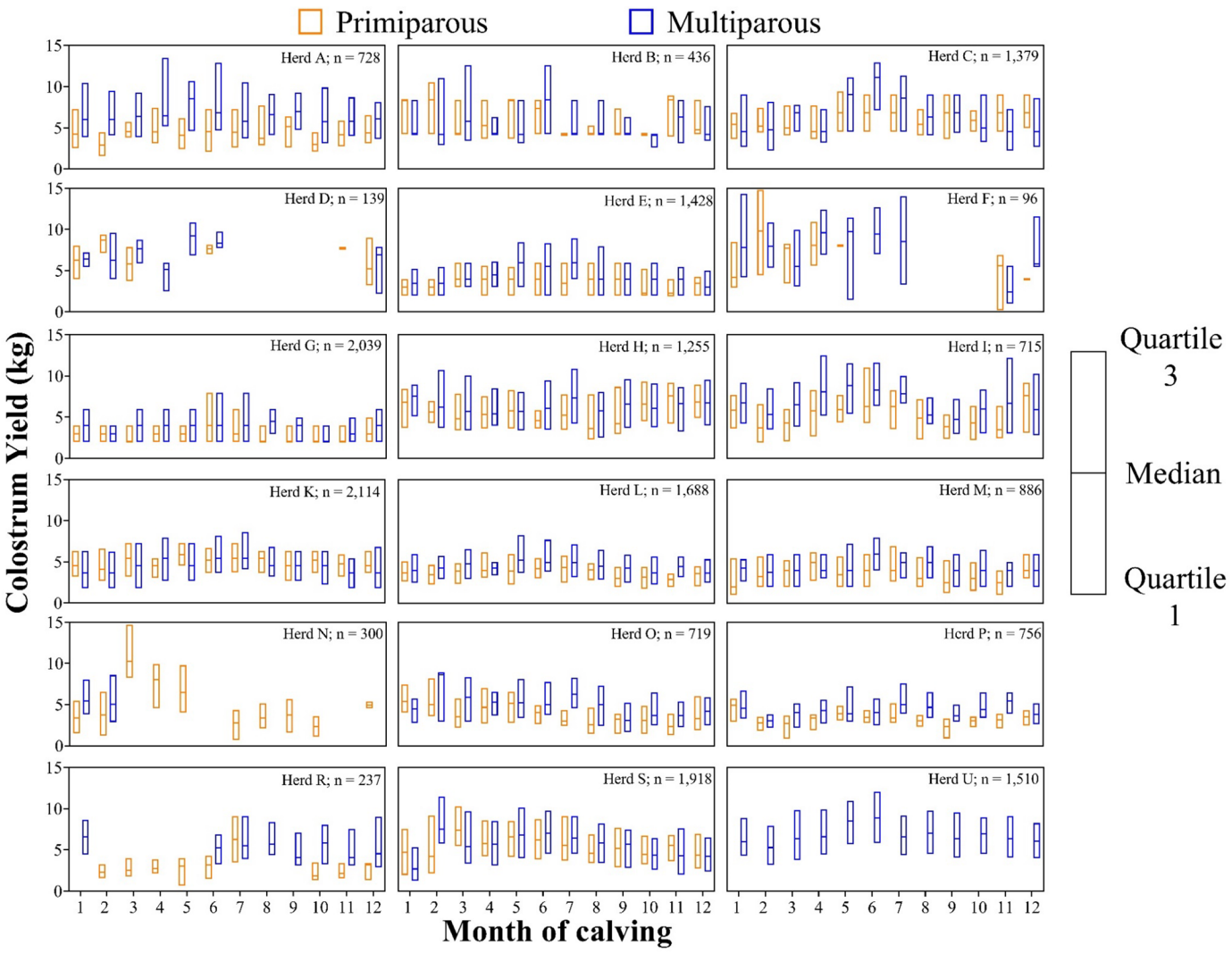


Figure 4. Box plots of monthly colostrum yield (kg) by enrolled farm from a convenience sample of 18 New York Holstein farms. Each panel contains herd code and sample size for the individual farm. Yellow and blue box plots represent primiparous and multiparous cows, respectively. Herd U housed 100% multiparous cows.

ing, the variables offered to the initial multivariable mixed model for colostrum yield from PP cows include Brix % (categorized), sex of the calf, age at first calving (categorized), and heat and humidity exposure AUC 7 d before calving (categorized). When controlling for month of calving ($P = 0.018$) and herd ($P = 0.004$), colostrum yield from PP cows was associated with Brix % ($P < 0.001$) and sex of the calf ($P = 0.008$). For colostrum Brix % from PP cows, sex of the calf, whether the calf was a stillbirth, light intensity AUC 14 d before calving (categorized), and DONCU diet (categorized) were offered to the initial multivariable mixed model after univariable screening. Colostrum Brix % from PP cows was associated with calf sex ($P = 0.003$), whether the calf was a stillbirth ($P = 0.01$), and light intensity AUC 14 d before calving ($P = 0.03$) when controlling for colostrum yield ($P = 0.098$), herd ($P = 0.009$), and month of calving ($P = 0.02$).

Multiparous Animals With Known Days on Close-up Diet

The initial model for colostrum yield from MPS cows after univariable screening included sex of the calf, whether the calf was a stillbirth, parity, Brix % (categorized), dry period length (categorized), previous lactation 305ME (categorized), gestation length (categorized), previous lactation length (categorized), heat and humidity exposure AUC 7 d before calving (categorized), light intensity AUC 14 d before calving

Table 5. Mixed effects multivariable models for variables associated with colostrum yield (kg) in Holstein primiparous (n = 5,790) and multiparous (n = 12,553) cows from 18 New York farms

	Primiparous		Multiparous	
Item	LSM[1] (95% CI)	*P*-value	LSM[1] (95% CI)	*P*-value
Brix %		<0.001		<0.001
≤22	3.9 (3.3–4.6)^{z}		6.0 (5.3–6.8)^{a}	
22.1–24.4	4.5 (3.8–5.3)^{xy}		5.7 (5.0–6.5)^{b}	
24.5–27	4.5 (3.8–5.3)^{x}		5.0 (4.4–5.7)^{c}	
≥27	4.2 (3.6–5.0)^{y}		4.1 (3.6–4.6)^{d}	
Calf sex		0.008		<0.001
Female	3.9 (3.4–4.5)^{y}		4.6 (4.0–5.2)^{c}	
Male	4.1 (3.5–4.8)^{x}		5.0 (4.4–5.7)^{b}	
Twin	4.9 (3.7–6.4)^{xy}		6.0 (5.2–7.8)^{a}	
Dry period length (d)				<0.001
<47			4.2 (3.7–4.7)^{c}	
47–67			5.1 (4.5–5.7)^{b}	
>67			6.5 (5.7–7.4)^{a}	
Previous lactation 305ME[2] (kg)				<0.001
≤13,090			4.9 (4.4–5.6)^{b}	
13,091–15,862			5.2 (4.6–5.9)^{a}	
>15,862			5.3 (4.7–6.0)^{a}	
Gestation length (d)				<0.001
263–273			4.8 (4.2–5.4)^{c}	
274–282			5.2 (4.6–5.8)^{b}	
283–293			5.5 (4.9–6.3)^{a}	
Stillbirth				0.006
Alive			5.5 (4.9–6.1)^{a}	
Dead			4.9 (4.2–5.6)^{b}	
Previous lactation length (d)				<0.001
<297			4.9 (4.3–5.5)^{b}	
297–344			5.0 (4.4–5.7)^{b}	
>344			5.6 (4.9–6.3)^{a}	
Parity				<0.001
2			5.4 (4.8–6.2)^{a}	
3			5.2 (4.6–5.9)^{b}	
4			4.9 (4.3–5.5)^{c}	
5+			5.0 (4.4–5.7)^{bc}	
Heat and humidity exposure				<0.001
AUC 7 d before calving[3] (average THI per 30-min interval)				
≤40.2			4.7 (4.1–5.4)^{c}	
40.3–50.1			4.9 (4.3–5.5)^{c}	
50.2–60.0			5.1 (4.5–5.8)^{bc}	
60.1–69.2			5.4 (4.7–6.1)^{ab}	
>69.2			5.7 (5.0–6.5)^{a}	
Light intensity AUC				0.012
14 d before calving[4] (average lux per 15-min interval)				
≤64.0			5.0 (4.4–5.7)^{ab}	
64.1–154.2			5.0 (4.5–5.7)^{b}	
>154.2			5.4 (4.7–6.1)^{a}	

$^{x–z}$Back-transformed LSM (95% CI) with different superscripts differ ($P < 0.05$; Tukey's test).

$^{a–d}$Back-transformed LSM (95% CI) with different superscripts differ ($P < 0.05$; Tukey's test).

[1]Data natural logarithm transformed before analysis. Models included random effects of herd and month of calving.

[2]305ME = 305-d mature equivalent milk yield.

[3]Area under the curve (AUC) was calculated for temperature-humidity index (THI) 7 d before calving in 30-min intervals. THI was collected from the close-up dry cow pen at each farm.

[4]Area under the curve was calculated for light intensity (lux) 14 d before calving in 15-min intervals. Lux was collected from the close-up dry cow pen at each farm.

(categorized), and DONCU diet (categorized). Variables remaining in the final model for colostrum yield from MPS cows include colostrum Brix % ($P < 0.001$), calf sex ($P < 0.001$), dry period length ($P < 0.001$), previous lactation 305ME ($P < 0.001$), gestation length ($P < 0.001$), previous lactation length ($P < 0.001$), whether the calf was a stillbirth ($P < 0.001$), parity ($P < 0.001$), heat and humidity exposure AUC 7 d

Table 6. Mixed effects multivariable models for variables associated with colostrum Brix % in Holstein primiparous (n = 5,790) and multiparous (n = 12,553) cows from 18 New York farms

	Primiparous[1]		Multiparous	
Item	LSM[2] (95% CI)	*P*-value	LSM[2] (95% CI)	*P*-value
Calf sex		<0.001		
Female	24.2 (23.4–25.1)[y]			
Male	24.6 (23.8–25.5)[x]			
Twin	25.2 (23.7–26.7)[xy]			
Stillbirth		0.013		0.005
Alive	25.0 (24.1–25.9)[x]		26.2 (25.4–27.0)[a]	
Dead	24.4 (23.4–25.4)[y]		25.6 (24.6–26.5)[b]	
Dry period length (d)				<0.001
<47			25.6 (24.7–26.4)[b]	
47–67			25.7 (24.9–26.5)[b]	
>67			26.4 (25.6–27.3)[a]	
Previous lactation 305ME[3] (kg)				0.005
≤13,090			26.0 (25.2–26.9)[a]	
13,091–15,862			25.9 (25.1–26.8)[a]	
>15,862			25.7 (24.9–26.5)[b]	
Gestation length (d)				0.01
263–273			25.8 (25.0–26.7)[ab]	
274–282			26.0 (25.2–26.9)[a]	
283–293			25.8 (25.0–26.6)[b]	
Colostrum yield (kg)				<0.001
<6			26.7 (25.9–27.5)[a]	
≥6			25.1 (24.3–25.9)[b]	
Parity				<0.001
2			24.4 (23.6–25.2)[d]	
3			25.6 (24.7–26.4)[c]	
4			26.3 (25.5–27.2)[b]	
5+			27.3 (26.4–28.1)[a]	
Heat and humidity exposure				0.003
AUC 7 d before calving[4] (average THI per 30-min interval)				
≤40.2			26.3 (25.5–27.2)[a]	
40.3–50.1			26.2 (25.4–27.0)[a]	
50.2–60.0			25.7 (24.9–26.6)[b]	
60.1–69.2			25.7 (24.9–26.6)[ab]	
>69.2			25.5 (24.6–26.3)[b]	
Light intensity AUC		0.011		
14 d before calving[5] (average lux per 15-min interval)				
≤64.0	25.0 (24.0–25.9)[x]			
64.1–154.2	24.5 (23.6–25.5)[y]			
>154.2	24.6 (23.6–25.5)[xy]			

[x,y]LSM (95% CI) with different superscripts differ ($P < 0.05$; Tukey's test).

[a–d]LSM (95% CI) with different superscripts differ ($P < 0.05$; Tukey's test).

[1]Colostrum yield was included as a covariate.

[2]Models included random effects of herd and month of calving.

[3]305ME = 305-d mature equivalent milk yield.

[4]Area under the curve (AUC) was calculated for temperature-humidity index (THI) 7 d before calving in 30-min intervals. THI was collected from the close-up dry cow pen at each farm.

[5]Area under the curve was calculated for light intensity (lux) 14 d before calving in 15-min intervals. Lux was collected from the close-up dry cow pen at each farm.

before calving ($P = 0.002$), and light intensity AUC 14 d before calving ($P = 0.002$) when controlling for herd ($P = 0.004$) and month of calving ($P = 0.018$). For colostrum Brix %, colostrum yield (categorized), sex of the calf, parity, whether the calf was a stillbirth, dry period length (categorized), previous lactation 305ME (categorized), gestation length (categorized), previous lactation length (categorized), and heat and humidity exposure AUC 7 d before calving (categorized) were offered to the initial multivariable mixed model after univariable screening. Colostrum Brix % from MPS cows was associated with colostrum yield ($P < 0.001$), dry period length ($P < 0.001$), previous lactation 305ME ($P < 0.001$), gestation length ($P = 0.005$), whether the

Table 7. Descriptive statistics of continuous variables for 923 cows with a 0-kg recorded colostrum yield and 18,343 cows with >0 kg recorded colostrum yield from 18 New York Holstein dairy farms

Variable	Mean	Median	SD	Range
Gestation length (d)				
Cows with 0-kg recorded colostrum yield	277.78	277	4.73	261–293
Cows with >0-kg recorded colostrum yield	277.28	277	4.93	261–293
Age at first calving[1] (mo)				
Cows with 0-kg recorded colostrum yield	21.72	21	1.22	19–27
Cows with >0-kg recorded colostrum yield	21.90	22	1.46	17–39
Dry period length (d)				
Cows with 0-kg recorded colostrum yield	53.42	53	7.35	27–97
Cows with >0-kg recorded colostrum yield	57.24	56	10.81	20–161
Previous lactation length (d)				
Cows with 0-kg recorded colostrum yield	325.82	308	39.96	270–550
Cows with >0-kg recorded colostrum yield	324.56	308	45.27	204–652
Previous lactation 305ME[2] (kg)				
Cows with 0-kg recorded colostrum yield	14,285	14,269	2,106	5,853–24,310
Cows with >0-kg recorded colostrum yield	14,500	14,469	2,167	5,416–29,493

[1]Only primiparous cows were included.
[2]305ME = 305-d mature equivalent milk yield.

calf was a stillbirth ($P = 0.003$), parity ($P < 0.001$), and heat and humidity exposure AUC 7 d before calving ($P = 0.008$) when controlling for herd ($P = 0.003$), and month of calving ($P = 0.02$).

Cows With 0 kg Recorded Colostrum Yield

A proportion of PP (n = 188, 3.1%) and MPS (n = 735, 5.5%) cows had a 0 kg recorded colostrum yield. Primiparous cows carrying twins were removed from further analysis due to a limited number with a 0-kg recorded colostrum yield (n = 1). Descriptive statistics for cows with a 0-kg and >0-kg recorded colostrum yield are presented in Tables 7 and 8. After univariable screening, the variables offered to the initial logistic regression model for PP cows included sex of the calf, whether the calf was a stillbirth, heat and humidity exposure AUC 7 d before calving (categorized), and month of calving. In the final model (Table 9), a 0-kg recorded colostrum yield from PP cows was associated with sex of the calf ($P = 0.004$), a stillbirth ($P = 0.048$), and month of calving ($P < 0.001$) when controlling for herd ($P = 0.017$). For MPS cows, sex of the calf, parity, dry period length (categorized), previous lactation 305ME (categorized), month of calving, previous lactation length (categorized), heat and humidity exposure AUC 7 d before calving (categorized), and light intensity AUC 14 d before calving (categorized) were offered to the initial logistic regression model after univariable screening. When controlling for herd ($P = 0.005$), sex of the calf ($P = 0.013$), parity ($P < 0.001$), dry period length ($P < 0.001$), previous lactation 305ME ($P = 0.048$), month of calving ($P = 0.003$), light intensity AUC 14 d before calving ($P = 0.089$), and heat and humidity exposure AUC 7 d before calving ($P = 0.006$) were associated with a 0-kg recorded colostrum yield from MPS cows (Table 10). Cows with a dry period ≤67 d and entering parity ≥3 had an increased odds of a 0 kg recorded colostrum yield.

DISCUSSION

One primary objective in this study was to describe colostrum production and colostrum handling practices on multiple New York farms. We found approximately one-fifth of cows produced colostrum below the industry standard (IgG ≥50 g/L or Brix % ≥22%). Gulliksen et al. (2008) reported 58% of 1,250 colostrum samples from Norwegian Red cows contained IgG <50 g/L. Other studies from Ireland (Barry et al., 2019) and the United States (Morrill et al., 2012; Quigley et al., 2013) found 21, 29, and 16%, respectively, of samples had IgG <50 g/L. Godden et al. (2019) suggest producers should strive for high-quality colostrum from ≥90% of animals. In the present study, 3 (16.7%) farms had high-quality (Brix ≥22%) colostrum in ≥90% of colostrum records.

Colostrum management programs are important in ensuring that clean, high-quality colostrum is delivered to the calf. On many farms in our study, colostrum was harvested within 8 h of calving. Kehoe et al. (2007) found that on 78.2% of the 55 Pennsylvania farms in their study, colostrum harvest was reported to occur within 6 h of calving. Timely colostrum harvest is important as previous research showed that IgG concentration decreased ≥9 h after calving (Conneely et al., 2013; Quigley et al., 2013). In our study, on 18 farms (94.7%) refrigerated or frozen stored colostrum was reported to be fed. This is different from studies con-

Table 8. Descriptive statistics of categorical variables for 923 cows with a 0-kg recorded colostrum yield and 18,343 cows with >0-kg recorded colostrum yield from 18 New York Holstein dairy farms

Variable	Cows with 0 kg of recorded colostrum yield (%)	Cows with >0 kg of recorded colostrum yield (%)
Parity		
1	21.2	31.6
2	26.2	28.0
3	23.4	20.3
4	13.4	11.4
5+	15.8	8.7
Month of calving		
1	10.0	10.8
2	8.6	6.8
3	6.2	8.0
4	5.8	7.2
5	6.3	7.6
6	3.9	8.6
7	3.7	7.9
8	5.7	8.9
9	9.8	8.0
10	12.6	8.2
11	11.6	7.8
12	15.8	10.2
Calf sex		
Female	59.0	53.2
Male	35.7	43.6
Twin	5.3	3.2
Stillbirth		
Alive	94.2	97.3
Dead	5.8	2.7
Heat and humidity exposure		
AUC 7 d before calving[1] (average THI per 30-min interval)		
≤40.2	11.4	6.0
40.3–50.1	40.8	39.1
50.2–60.0	19.6	19.5
60.1–69.2	18.2	20.7
>69.2	10.0	14.7
Light intensity AUC 14 d before calving[2] (average lux per 15-min interval)		
≤64.0	31.4	24.5
64.1–154.2	54.2	50.8
>154.2	14.4	24.7
Gestation length[3] (d)		
Short	16.1	18.3
Normal	66.0	63.8
Long	17.9	17.9
Dry period length (d)		
<47	14.0	9.7
47–67	82.6	79.6
>67	3.4	10.7
Previous lactation length (d)		
<297	14.7	26.8
297–344	60.8	48.5
>344	24.5	24.7
Previous lactation 305ME[4] (kg)		
≤13,090	29.5	25.0
13,091–15,862	48.3	50.0
>15,862	22.2	25.0

[1]Area under the curve (AUC) was calculated for temperature-humidity index (THI) 7 d before calving in 30-min intervals. THI was collected from the close-up dry cow pen at each farm.

[2]Area under the curve was calculated for light intensity (lux) 14 d before calving in 15-min intervals. Lux was collected from the close-up dry cow pen at each farm.

[3]Primiparous (PP) and multiparous (MPS) gestation length: short (PP = 261–271, MPS = 263–273 d), normal (PP = 272–280, MPS = 274–282 d), or long (PP = 281–291, MPS = 283–293 d).

[4]305ME = 305-d mature equivalent milk yield.

Table 9. Logistic regression analysis for variables associated with a 0-kg recorded colostrum yield in Holstein primiparous (n = 5,948) cows from a convenience sample of 18 New York dairy farms

Variable	Odds ratio	95% CI	*P*-value
Calf sex			0.004
Female	Referent		
Male	0.571	0.389–0.838	
Stillbirth			0.048
Alive	Referent		
Dead	1.943	1.007–3.749	
Month of calving			<0.001
1	Referent		
2	1.710	0.900–3.248	
3	0.962	0.477–1.943	
4	0.896	0.446–1.802	
5	0.537	0.224–1.286	
6	0.288	0.097–0.855	
7	0.351	0.130–0.951	
8	1.065	0.565–2.009	
9	1.626	0.898–2.947	
10	1.028	0.512–2.066	
11	0.795	0.364–1.735	
12	2.122	1.176–3.830	

ducted in eastern Canada and the Netherlands where most calves were fed fresh colostrum from their dam (Renaud et al., 2020, Robbers et al., 2021); although, farms in these studies had fewer cows than those in the current study. Kehoe et al. (2007) reported that on 89% of farms with ≤200 cows, calves were fed colostrum from their dam, whereas on larger farms (>200 cows), colostrum was often stored and only 43% fed fresh colostrum from the dam. On most farms in the present study, first colostrum was fed within 2 h after birth. This is in line with work from Robbers et al. (2021) where 84% of calves received colostrum within 2 h of birth. Feeding colostrum within 1 h of birth increased apparent efficiency of IgG absorption and 24 h serum IgG concentration compared with calves fed colostrum at 6 or 12 h (Fischer et al., 2018). Feeding a second meal of colostrum has been associated with reduced odds of poor transfer of passive immunity and morbidity, and tended to have greater first lactation 305ME (Abuelo et al., 2021). In the current study, a second colostrum meal was fed to heifer and bull calves on just over half and one-third of farms, respectively. This is greater than the 26% of Pennsylvania farms for which feeding a second colostrum meal was reported (Kehoe et al., 2007). Colostrum with high bacterial contamination decreased apparent efficiency of absorption and plasma IgG concentrations (Gelsinger et al., 2015). Few producers (n = 3; 15.8%) in the present study or previous work (n = 9; 8.7%; Urie et al., 2018) heat treat colostrum. Heat treating colostrum at 60°C is effective at lowering total bacteria count with minimal reduction in IgG concentration; however, the reduction in bacteria counts is affected by the bacteria group, as some bacteria are more tolerant to heat (Mann et al., 2020; Malik et al., 2022).

Another primary objective was to identify cow, management, and environmental factors associated with colostrum yield and Brix %. In agreement with other studies (Dunn et al., 2017; Gavin et al., 2018; Borchardt et al., 2022), our results show several factors were associated with colostrum yield and Brix %.

Cows having a female calf were associated with the lowest colostrum yield in the current study. These findings are in line with reports by other authors (Borchardt et al., 2022) who showed an increase in colostrum yield with male or twin calves compared with a female calf in MPS cows, but different from Angulo et al. (2015) who reported cows carrying a female calf produced more colostrum than those with a male. Calf birth weight was not available in the current study; however, in a previous study, colostrum yield was shown to increase as calf birth weight increased (Conneely et al., 2013). Greater birth weights are associated with longer gestation lengths and male calves, and higher birth weights were positively correlated with milk production and placental weight (Chew et al., 1981; Kertz et al., 1997; Atashi and Asaadi, 2019). Although the endocrine regulation of bovine colostrogenesis is not fully understood to date, hormones including bovine placental lactogen, cortisol, prolactin, and estrogen, some which are known to vary with placental weight (Chew et al., 1981) could be contributing to mammary development, colostrogenesis, and the observed association of short gestation lengths and reduced colostrum yield from MPS cows.

A stillbirth was associated with lower colostrum yield in MPS cows and decreased Brix % in the current study. Borchardt et al. (2022) reported colostrum yield was lower with a stillborn calf, and colostrum yield in sows was negatively correlated (r = −0.33) with the number of stillbirths (Quesnel, 2011). Patel et al. (1996) reported circulating cortisol at parturition was greater in a cow giving birth to stillborn calves compared with cows giving birth to calves that were alive. Other studies have associated stillborn calves and low calf viability with low blood concentrations of placental derived hormones (pregnancy-associated glycoproteins and estrone sulfate; Kindahl et al., 2004). We are unable to consider the time of death of the stillborn calves in relation to the parturition of a cow, as well as in relation to any calving difficulty which may have produced an acute stress response in the cow, interfering with successful harvest of the colostrum present in the mammary gland. Therefore, both changes in the endocrine status during late gestation or parturition, altering colostrogenesis, as well as inhibition of the milk letdown reflex, altering colostrum harvest, could play a role in the observed association.

Table 10. Logistic regression analysis for variables associated with a 0-kg recorded colostrum yield in Holstein multiparous (n = 13,288) cows from a convenience sample of 18 New York dairy farms

Variable	Odds ratio	95% CI	*P*-value
Calf sex			0.013
Female	Referent		
Male	0.778	0.658–0.919	
Twin	0.882	0.606–1.282	
Parity			<0.001
2	Referent		
3	1.296	1.058–1.587	
4	1.391	1.096–1.765	
5+	2.207	1.751–2.784	
Dry period length (d)			<0.001
<47	3.468	2.141–5.619	
47–67	2.014	1.307–3.101	
>67	Referent		
Previous lactation 305ME[1] (kg)			0.048
≤13,090	Referent		
13,091–15,862	0.850	0.702–1.029	
>15,862	0.749	0.593–0.946	
Month of calving			0.003
1	Referent		
2	1.175	0.746–1.853	
3	0.746	0.472–1.179	
4	0.826	0.517–1.320	
5	0.970	0.586–1.603	
6	0.498	0.258–0.962	
7	0.415	0.198–0.872	
8	0.449	0.231–0.871	
9	0.994	0.568–1.738	
10	1.427	0.880–2.313	
11	1.177	0.815–1.698	
12	1.106	0.795–1.540	
Light intensity AUC			0.089
14 d before calving[2] (average lux per 15-min interval)			
≤64.0	Referent		
64.1–154.2	0.780	0.585–1.040	
>154.2	0.550	0.323–0.938	
Heat and humidity exposure			0.006
AUC 7 d before calving[3] (average THI per 30-min interval)			
≤40.2	Referent		
40.3–50.1	1.042	0.741–1.467	
50.2–60.0	0.626	0.390–1.006	
60.1–69.2	0.937	0.534–1.645	
>69.2	1.278	0.649–2.517	

[1]305ME = 305-d mature equivalent milk yield.

[2]Area under the curve (AUC) was calculated for light intensity (lux) 14 d before calving in 15-min intervals. Lux was collected from the close-up dry cow pen at each farm.

[3]Area under the curve was calculated for temperature-humidity index (THI) 7 d before calving in 30-min intervals. THI was collected from the close-up dry cow pen at each farm.

Shortening without omitting the dry period has been reported to decrease colostrum yield but not quality (Mansfeld et al., 2012; Mayasari et al., 2015). Colostrum yield increased as dry period length increased; however, Brix % was not different for animals with <47 or 47 to 67 d dry period in the current study. The IgG concentration in prepartum secretions from continuously milked cows remained stable from approximately −5 d to parturition (Baumrucker et al., 2014), indicating that transfer of IgG into the mammary gland can occur quickly during the last days of gestation. Colostrum yield, however, was reduced in animals assigned to a 40-d dry period compared with those with a 60-d dry period (Grusenmeyer et al., 2006). Similarly, milk yield in the subsequent lactation was decreased with a shortened (34 and 35 d) dry period (Watters et al., 2008; Bernier-Dodier et al., 2011). Our data also suggest a shorter dry period length of ≤67 d increased the odds of a 0-kg recorded colostrum yield by 2.01 to 3.47 times compared with cows with >67 d dry period. Most mammary cell proliferation occurs during the dry period (Sorensen et al., 2006); therefore, reduced dry

period length could affect cell proliferation or function during colostrogenesis.

Colostrum IgG concentration was highest in parity ≥3 compared with parity 1 or 2 (Angulo et al., 2015; Bartier et al., 2015; Shivley et al., 2018). In the current study, colostrum Brix % increased as parity increased. Lower serum IgG concentrations have been reported in younger cows (Devery-Pocius and Larson, 1983). Conneely et al. (2013) found colostrum yield to be the highest in parity 2, 3, or 4 and lowest in parity 1 animals, but in other studies (Kehoe et al., 2011; Borchardt et al., 2022), colostrum yield was not associated with parity. In the current study, cows entering parity 2 were associated with the greatest colostrum yield. Gavin et al. (2018) reported no colostrum was produced in 0.3% of PP and 6.0% of MPS cows. Although, in our study, the proportion of PP animals with 0-kg recorded colostrum yield was higher (3.1%) than that reported by Gavin et al. (2018), the proportion of MPS animals are very similar (5.5%).

In the present study, previous lactation 305ME >15,862 and >13,090 kg were associated with lowest Brix % and greatest colostrum yield, respectively, although the changes are relatively small. Colostrum IgG concentration was weakly positively correlated with previous lactation milk production (r = 0.07–0.30) and DIM (r = 0.21) in 2 earlier studies (Pritchett et al., 1991; Cabral et al., 2016). In the current study, previous lactation length was not associated with Brix %, but colostrum yield was greatest with a previous lactation length >344 d. Borchardt et al. (2022) found previous lactation 305-d milk yield to be positively and negatively associated with colostrum yield and Brix %, respectively, whereas in other studies, there was no association with IgG concentration or colostrum yield (Dunn et al., 2017; Gavin et al., 2018).

Annual rhythms in milk production and components in different regions of the United States exist and appear to be consistent between years (Salfer and Harvatine, 2018); however, limited data are available investigating seasonal variation in colostrum production. Borchardt et al. (2022) reported colostrum yield and Brix % were lowest in November and August in PP, and in October and August in MPS cows, respectively. Other studies, however, have shown colostrum quality not to be associated with season or month of calving (Pritchett et al., 1991; Bartier et al., 2015; Dunn et al., 2017). In the current study, median colostrum yield and average Brix % were lowest in October and May in PP cows and in February and June in MPS cows, respectively. Gavin et al. (2018) found that approximately 35% of MPS cows were not producing colostrum in December. In our study, 2.9 and 9.5% of records from MPS cows had a 0-kg recorded colostrum yield in June and December, respectively.

Light intensity and heat and humidity exposure during the final weeks of gestation, controlling for month of calving and herd, were investigated in our present study to account for short-term changes in these parameters that would not be adequately expressed by season or even month of the year. Colostrum yield and Brix % from MPS cows were associated with heat and humidity exposure AUC 7 d before calving in the present study. The number of days >23°C in the last 21 d of gestation was weakly positively correlated (r = 0.3) with colostrum yield, whereas correlation with IgG concentration was negative (r = −0.24; Cabral et al., 2016). Gavin et al. (2018) reported colostrum production in a Jersey herd over a 1-yr period and showed that colostrum yield followed a similar pattern as average maximum THI and photoperiod 1 mo before calving. In addition, THI and photoperiod had a stronger correlation with colostrum yield in MPS cows than it did in PP cows, and the authors speculate that the endocrine response to photoperiod could have a role in colostrogenesis (Gavin et al., 2018). In the current study MPS cows exposed to an average of >154.2 lux per 15-min interval 14 d before calving were associated with greater colostrum yield and had 0.45 times the odds of a 0-kg recorded colostrum yield compared with ≤64.0 lux per 15-min interval 14 d before calving. Short day exposure (8 h light; 16 h dark) during the dry period decreased the prolactin surge and increased milk yield through 120 DIM (Auchtung et al., 2005; Velasco et al., 2008), but did not affect colostrum IgG concentration and yield (Morin et al., 2010).

In an analysis of a subset of data, DONCU was not associated with colostrum yield or Brix % in the current study. Borchardt et al. (2022) reported no association with days in the close-up group and colostrum yield from PP cows or Brix %; however, a positive association was found for colostrum yield from MPS cow. Furthermore, MPS cows assigned to a 21-d close-up period produced numerically greater colostrum (8.80 vs. 6.75 kg) compared with cows assigned to a 10-d close-up period (Farahani et al., 2017). The authors speculate that inadequate time on the close-up diet might not meet the increased nutrient demands of late gestation and colostrogenesis (Borchardt et al., 2022); however, content of the close-up diet was not evaluated in this analysis and warrants further investigation.

Although future studies are needed to determine the cause-effect relationship between the observed variables that were associated with changes in colostrum yield and Brix %, recognizing these factors associated with colostrum production and altering colostrum manage-

ment programs remains a strategic opportunity for farms struggling with periods of low colostrum supply. In agreement with the present study, other authors have reported reduced colostrum yield when shortening the dry period (Mansfeld et al., 2012; Mayasari et al., 2015; O'Hara et al., 2019). Additionally, other factors should be considered. Colostrum bacterial contamination has been reported to decrease apparent efficiency of IgG absorption and plasma IgG concentration (Gelsinger et al., 2015); therefore, cleanliness of equipment and proper cooling and storage methods could maximize the efficiency of use of available colostrum supply. Transition milk (milking 2 to 6) contains greater concentration of important components compared with mature milk (Godden et al., 2019). Implementing transition milk after colostrum feeding has shown benefits to calf health and intestinal maturation (Fischer et al., 2019).

We have some limitations to consider with the current study. Enrolled farms were a convenience sample of mid- to large-size New York Holstein dairy farms where herd personnel had the ability and willingness to collect and record colostrum yield and Brix % for our study. Due to our enrollment criteria, the average (range) herd size of 1,545 (620 to 4,600) cows is larger than the average herd size of 172 cows in New York (American Dairy Association Northeast, 2021). The results from the current study are relevant to farms with similar management practices, and the number of farms and observations increases external validity of the reported findings. However, caution should be taken when translating the results to farms with different farm management strategies or in other geographic regions. With only 19 farms enrolled, the reported colostrum management practices could misrepresent those of other similarly sized farms that are not included, as well as those practiced on smaller farms. After initial training, farm personnel were responsible for colostrum collection and recording yield and quality; therefore, differences in colostrum collection and data recording could be present. In addition, the length of colostrum record collection and percent of calvings with a recorded final colostrum record among all calvings differed by farm. Due to the large sample size and associated prohibitive costs, the gold standard for colostrum quality, radial immunodiffusion, was not used in this study; however, Brix % is an acceptable estimate for on farm use (Godden et al., 2019). The possible biological significance of small differences in yield and Brix % determined in our statistical analysis should also be considered.

CONCLUSIONS

We described colostrum management practices from multiple New York Holstein herds as well as their monthly colostrum production. When controlling for the effect of month of calving and the effect of herd, colostrum yield and Brix % were associated with prepartum environmental conditions, cow, and management factors. Recognition of factors associated with the variation in colostrum production will allow producers to alter their colostrum management programs during periods of low yield or quality for an opportunity to decrease calf morbidity or mortality.

ACKNOWLEDGMENTS

This study was supported in part by the New York Farm Viability Institute (Syracuse, NY). We thank the participating dairy producers and their staff for their willingness to be included in the study. The authors have not stated any conflicts of interest.

REFERENCES

Abuelo, A., F. Cullens, A. Hanes, and J. L. Brester. 2021. Impact of 2 versus 1 colostrum meals on failure of transfer of passive immunity, pre-weaning morbidity and mortality, and performance of dairy calves in a large dairy herd. Animals (Basel) 11:782. https://doi.org/10.3390/ani11030782.

Ahmann, J., J. Steinhoff-Wagner, and W. Buscher. 2021. Determining immunoglobulin content of bovine colostrum and factors affecting the outcome: A review. Animals (Basel) 11:3587. https://doi.org/10.3390/ani11123587.

American Dairy Association Northeast. 2021. Dairy Stats by State. Accessed Apr. 26, 2022. https://www.americandairy.com/dairy-farms/state-stats/.

Angulo, J., L. M. Gomez, L. Mahecha, E. Mejia, J. Henao, and C. Mesa. 2015. Calf's sex, parity and the hour of harvest after calving affect colostrum quality of dairy cows grazing under high tropical conditions. Trop. Anim. Health Prod. 47:699–705. https://doi.org/10.1007/s11250-015-0781-z.

Atashi, H., and A. Asaadi. 2019. Association between gestation length and lactation performance, lactation curve, calf birth weight and dystocia in Holstein dairy cows in Iran. Anim. Reprod. 16:846–852. https://doi.org/10.21451/1984-3143-AR2019-0005.

Auchtung, T. L., A. G. Rius, P. E. Kendall, T. B. McFadden, and G. E. Dahl. 2005. Effects of photoperiod during the dry period on prolactin, prolactin receptor, and milk production of dairy cows. J. Dairy Sci. 88:121–127. https://doi.org/10.3168/jds.S0022-0302(05)72669-2.

Barry, J., E. A. Bokkers, D. Berry, I. J. de Boer, J. McClure, and E. Kennedy. 2019. Associations between colostrum management, passive immunity, calf-related hygiene practices, and rates of mortality in preweaning dairy calves. J. Dairy Sci. 102:10266–10276. https://doi.org/10.3168/jds.2019-16815.

Bartier, A. L., M. C. Windeyer, and L. Doepel. 2015. Evaluation of on-farm tools for colostrum quality measurement. J. Dairy Sci. 98:1878–1884. https://doi.org/10.3168/jds.2014-8415.

Baumrucker, C. R., R. S. Zbinden, H. A. van Dorland, G. J. Remmelink, B. Kemp, A. T. van Knegsel, and R. M. Bruckmaier. 2014. Continuous milking of dairy cows disrupts timing of peak IgG concentration appearance in mammary secretions. J. Dairy Res. 81:403–409. https://doi.org/10.1017/S002202991400034X.

Bernier-Dodier, P., C. L. Girard, B. G. Talbot, and P. Lacasse. 2011. Effect of dry period management on mammary gland function and its endocrine regulation in dairy cows. J. Dairy Sci. 94:4922–4936. https://doi.org/10.3168/jds.2010-4116.

Bielmann, V., J. Gillan, N. R. Perkins, A. L. Skidmore, S. Godden, and K. E. Leslie. 2010. An evaluation of Brix refractometry instru-

ments for measurement of colostrum quality in dairy cattle. J. Dairy Sci. 93:3713–3721. https://doi.org/10.3168/jds.2009-2943.
Borchardt, S., F. Sutter, W. Heuwieser, and P. Venjakob. 2022. Management-related factors in dry cows and their associations with colostrum quantity and quality on a large commercial dairy farm. J. Dairy Sci. 105:1589–1602. https://doi.org/10.3168/jds.2021-20671.
Cabral, R. G., C. E. Chapman, K. M. Aragona, E. Clark, M. Lunak, and P. S. Erickson. 2016. Predicting colostrum quality from performance in the previous lactation and environmental changes. J. Dairy Sci. 99:4048–4055. https://doi.org/10.3168/jds.2015-9868.
Chew, B., L. Maier, J. Hillers, and A. Hodgson. 1981. Relationship between calf birth weight and dam's subsequent 200-and 305-day yields of milk, fat, and total solids in Holsteins. J. Dairy Sci. 64:2401–2408. https://doi.org/10.3168/jds.S0022-0302(81)82863-9.
Conneely, M., D. P. Berry, R. Sayers, J. P. Murphy, I. Lorenz, M. L. Doherty, and E. Kennedy. 2013. Factors associated with the concentration of immunoglobulin G in the colostrum of dairy cows. Animal 7:1824–1832. https://doi.org/10.1017/S1751731113001444.
Devery-Pocius, J. E., and B. L. Larson. 1983. Age and previous lactations as factors in the amount of bovine colostral immunoglobulins. J. Dairy Sci. 66:221–226. https://doi.org/10.3168/jds.S0022-0302(83)81780-9.
Dunn, A., A. Ashfield, B. Earley, M. Welsh, A. Gordon, and S. J. Morrison. 2017. Evaluation of factors associated with immunoglobulin G, fat, protein, and lactose concentrations in bovine colostrum and colostrum management practices in grassland-based dairy systems in Northern Ireland. J. Dairy Sci. 100:2068–2079. https://doi.org/10.3168/jds.2016-11724.
Farahani, T. A., H. Amanlou, and M. Kazemi-Bonchenari. 2017. Effects of shortening the close-up period length coupled with increased supply of metabolizable protein on performance and metabolic status of multiparous Holstein cows. J. Dairy Sci. 100:6199–6217. https://doi.org/10.3168/jds.2016-12263.
Fischer, A., C. Villot, J. van Niekerk, T. Yohe, D. Renaud, and M. Steele. 2019. Invited Review: Nutritional regulation of gut function in dairy calves: From colostrum to weaning. Appl. Anim. Sci. 35:498–510. https://doi.org/10.15232/aas.2019-01887.
Fischer, A. J., Y. Song, Z. He, D. M. Haines, L. L. Guan, and M. A. Steele. 2018. Effect of delaying colostrum feeding on passive transfer and intestinal bacterial colonization in neonatal male Holstein calves. J. Dairy Sci. 101:3099–3109. https://doi.org/10.3168/jds.2017-13397.
Fischer-Tlustos, A. J., A. Lopez, K. S. Hare, K. M. Wood, and M. A. Steele. 2021. Invited Review: Effects of colostrum management on transfer of passive immunity and the potential role of colostral bioactive components on neonatal calf development and metabolism. Can. J. Anim. Sci. 101:405–426. https://doi.org/10.1139/cjas-2020-0149.
Gavin, K., H. Neibergs, A. Hoffman, J. N. Kiser, M. A. Cornmesser, S. A. Haredasht, B. Martinez-Lopez, J. R. Wenz, and D. A. Moore. 2018. Low colostrum yield in Jersey cattle and potential risk factors. J. Dairy Sci. 101:6388–6398. https://doi.org/10.3168/jds.2017-14308.
Gelsinger, S. L., C. M. Jones, and A. J. Heinrichs. 2015. Effect of colostrum heat treatment and bacterial population on immunoglobulin G absorption and health of neonatal calves. J. Dairy Sci. 98:4640–4645. https://doi.org/10.3168/jds.2014-8790.
Godden, S. 2008. Colostrum management for dairy calves. Vet. Clin. North Am. Food Anim. Pract. 24:19–39. https://doi.org/10.1016/j.cvfa.2007.10.005.
Godden, S. M., J. E. Lombard, and A. R. Woolums. 2019. Colostrum management for dairy calves. Vet. Clin. North Am. Food Anim. Pract. 35:535–556. https://doi.org/10.1016/j.cvfa.2019.07.005.
Grusenmeyer, D., C. Ryan, D. Galton, and T. Overton. 2006. Shortening the dry period from 60 to 40 days does not affect colostrum quality but decreases colostrum yield by Holstein cows. J. Dairy Sci. 8(Suppl. 1):336. (Abstr.)
Gulliksen, S. M., K. I. Lie, L. Solverod, and O. Osteras. 2008. Risk factors associated with colostrum quality in Norwegian dairy cows. J. Dairy Sci. 91:704–712. https://doi.org/10.3168/jds.2007-0450.
Kehoe, S., A. J. Heinrichs, M. Moody, C. Jones, and M. Long. 2011. Comparison of immunoglobulin G concentrations in primiparous and multiparous bovine colostrum. Prof. Anim. Sci. 27:176–180. https://doi.org/10.15232/S1080-7446(15)30471-X.
Kehoe, S. I., B. M. Jayarao, and A. J. Heinrichs. 2007. A survey of bovine colostrum composition and colostrum management practices on Pennsylvania dairy farms. J. Dairy Sci. 90:4108–4116. https://doi.org/10.3168/jds.2007-0040.
Kertz, A. F., L. F. Reutzel, B. A. Barton, and R. L. Ely. 1997. Body weight, body condition score, and wither height of prepartum Holstein cows and birth weight and sex of calves by parity: A database and summary. J. Dairy Sci. 80:525–529. https://doi.org/10.3168/jds.S0022-0302(97)75966-6.
Kindahl, H., B. Kornmatitsuk, and H. Gustafsson. 2004. The cow in endocrine focus before and after calving. Reprod. Domest. Anim. 39:217–221. https://doi.org/10.1111/j.1439-0531.2004.00506.x.
Lopez, A. J., and A. J. Heinrichs. 2022. Invited review: The importance of colostrum in the newborn dairy calf. J. Dairy Sci. 105:2733–2749. https://doi.org/10.3168/jds.2020-20114.
Malik, M. I., M. A. Rashid, and D. Raboisson. 2022. Heat treatment of colostrum at 60°C decreases colostrum immunoglobulins but increases serum immunoglobulins and serum total protein: A meta-analysis. J. Dairy Sci. 105:3453–3467. https://doi.org/10.3168/jds.2021-21231.
Mann, S., G. Curone, T. Chandler, P. Moroni, J. Cha, R. Bhawal, and S. Zhang. 2020. Heat treatment of bovine colostrum: I. Effects on bacterial and somatic cell counts, immunoglobulin, insulin, and IGF-I concentrations, as well as the colostrum proteome. J. Dairy Sci. 103:9368–9383. https://doi.org/10.3168/jds.2020-18618.
Mansfeld, R., C. Sauter-Louis, and R. Martin. 2012. Auswirkungen der Länge der Trockenstehzeit bei Milchkühen auf Leistung, Gesundheit, Fruchtbarkeit und Kolostrumqualität. Tierarztl. Prax. Ausg. G Grosstiere Nutztiere 40:239–250. https://doi.org/10.1055/s-0038-1623125.
Mayasari, N., G. de Vries Reilingh, M. G. Nieuwland, G. J. Remmelink, H. K. Parmentier, B. Kemp, and A. T. van Knegsel. 2015. Effect of maternal dry period length on colostrum immunoglobulin content and on natural and specific antibody titers in calves. J. Dairy Sci. 98:3969–3979. https://doi.org/10.3168/jds.2014-8753.
Morin, D. E., P. D. Constable, F. P. Maunsell, and G. C. McCoy. 2001. Factors associated with colostral specific gravity in dairy cows. J. Dairy Sci. 84:937–943. https://doi.org/10.3168/jds.S0022-0302(01)74551-1.
Morin, D. E., S. V. Nelson, E. D. Reid, D. W. Nagy, G. E. Dahl, and P. D. Constable. 2010. Effect of colostral volume, interval between calving and first milking, and photoperiod on colostral IgG concentrations in dairy cows. J. Am. Vet. Med. Assoc. 237:420–428. https://doi.org/10.2460/javma.237.4.420.
Morrill, K. M., E. Conrad, A. Lago, J. Campbell, J. Quigley, and H. Tyler. 2012. Nationwide evaluation of quality and composition of colostrum on dairy farms in the United States. J. Dairy Sci. 95:3997–4005. https://doi.org/10.3168/jds.2011-5174.
Norman, H. D., J. R. Wright, M. T. Kuhn, S. M. Hubbard, J. B. Cole, and P. M. VanRaden. 2009. Genetic and environmental factors that affect gestation length in dairy cattle. J. Dairy Sci. 92:2259–2269. https://doi.org/10.3168/jds.2007-0982.
O'Hara, E. A., R. Båge, U. Emanuelson, and K. Holtenius. 2019. Effects of dry period length on metabolic status, fertility, udder health, and colostrum production in 2 cow breeds. J. Dairy Sci. 102:595–606. https://doi.org/10.3168/jds.2018-14873.
Patel, O. V., T. Takahashi, N. Takenouchi, M. Hirako, N. Sasaki, and I. Domekis. 1996. Peripheral cortisol levels throughoutgestation in the cow: Effect of stage of gestation and foetal number. Br. Vet. J. 152:425–432. https://doi.org/10.1016/S0007-1935(96)80036-4.
Pritchett, L. C., C. C. Gay, T. E. Besser, and D. D. Hancock. 1991. Management and production factors influencing immunoglobulin G1 concentration in colostrum from Holstein cows. J. Dairy Sci. 74:2336–2341. https://doi.org/10.3168/jds.S0022-0302(91)78406-3.

Quesnel, H. 2011. Colostrum production by sows: Variability of colostrum yield and immunoglobulin G concentrations. Animal 5:1546–1553. https://doi.org/10.1017/S175173111100070X.

Quigley, J. D., A. Lago, C. Chapman, P. Erickson, and J. Polo. 2013. Evaluation of the Brix refractometer to estimate immunoglobulin G concentration in bovine colostrum. J. Dairy Sci. 96:1148–1155. https://doi.org/10.3168/jds.2012-5823.

Renaud, D. L., M. A. Steele, R. Genore, S. M. Roche, and C. B. Winder. 2020. Passive immunity and colostrum management practices on Ontario dairy farms and auction facilities: A cross-sectional study. J. Dairy Sci. 103:8369–8377. https://doi.org/10.3168/jds.2020-18572.

Robbers, L., H. J. C. Bijkerk, A. P. Koets, L. Benedictus, M. Nielen, and R. Jorritsma. 2021. Survey on colostrum management by dairy farmers in the Netherlands. Front. Vet. Sci. 8:656391. https://doi.org/10.3389/fvets.2021.656391.

Salfer, I. J., and K. J. Harvatine. 2018. Seasonal variation in milk composition. Pages 157–166 in Tri-State Dairy Nutr. Conf. Fort Wayne, IN. Ohio State Univ.

Shivley, C. B., J. E. Lombard, N. J. Urie, D. M. Haines, R. Sargent, C. A. Kopral, T. J. Earleywine, J. D. Olson, and F. B. Garry. 2018. Preweaned heifer management on US dairy operations: Part II. Factors associated with colostrum quality and passive transfer status of dairy heifer calves. J. Dairy Sci. 101:9185–9198. https://doi.org/10.3168/jds.2017-14008.

Sorensen, M. T., J. V. Norgaard, P. K. Theil, M. Vestergaard, and K. Sejrsen. 2006. Cell turnover and activity in mammary tissue during lactation and the dry period in dairy cows. J. Dairy Sci. 89:4632–4639. https://doi.org/10.3168/jds.S0022-0302(06)72513-9.

Urie, N. J., J. E. Lombard, C. B. Shivley, C. A. Kopral, A. E. Adams, T. J. Earleywine, J. D. Olson, and F. B. Garry. 2018. Preweaned heifer management on US dairy operations: Part I. Descriptive characteristics of preweaned heifer raising practices. J. Dairy Sci. 101:9168–9184. https://doi.org/10.3168/jds.2017-14010.

Velasco, J. M., E. D. Reid, K. K. Fried, T. F. Gressley, R. L. Wallace, and G. E. Dahl. 2008. Short-day photoperiod increases milk yield in cows with a reduced dry period length. J. Dairy Sci. 91:3467–3473. https://doi.org/10.3168/jds.2008-1028.

Watters, R. D., J. N. Guenther, A. E. Brickner, R. R. Rastani, P. M. Crump, P. W. Clark, and R. R. Grummer. 2008. Effects of dry period length on milk production and health of dairy cattle. J. Dairy Sci. 91:2595–2603. https://doi.org/10.3168/jds.2007-0615.

Westhoff, T. A., S. J. Womack, T. R. Overton, C. M. Ryan, and S. Mann. 2022. Supplemental File from: Epidemiology of bovine colostrum production in New York Holstein herds: Cow, management, and environmental factors. eCommons. https://hdl.handle.net/1813/112245.

ORCIDS

T. A. Westhoff https://orcid.org/0000-0001-5989-0229
S. J. Womack https://orcid.org/0000-0003-4417-817X
T. R. Overton https://orcid.org/0000-0003-4098-4916
S. Mann https://orcid.org/0000-0003-1806-1154